

Valerio Valente

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 3 years of experience shipping AI agents, RAG systems and MLOps pipelines for pharma, manufacturing and research clients. Currently at Aixelerate, where I own the AI layer of a research platform - building agent-driven analysis tools, explainable statistical components and the product surfaces around them. Prior work: regulatory-compliant RAG (top 5 FDA challenge), tabular synthetic data, reproducible Azure ML pipelines.

CORE COMPETENCIES

AI Agents & Multi-Agent Systems

Retrieval-Augmented Generation (RAG)

Model Context Protocol (MCP)

LLM Evaluation & Observability

MLOps & ML Pipelines

Explainable AI

Production Python (FastAPI)

Cloud (Azure ML, AWS)

WORK EXPERIENCE

Aixelerate s.r.l. April 2026 - current

Data Scientist - AI & Product

- Built and shipped a suite of statistical-analysis components for clinical datasets, running end-to-end from input validation to user-facing outputs.
- Own the evolution of the AI layer toward a production multi-agent architecture, moving the system from development to production-grade.
- Authored Claude Code skills that standardize quality and cut repetitive scaffolding work.
- Contributed across the product stack with a strong UX focus, pushing usability improvements informed by researcher interviews.

InSilicoTrials Technologies SpA April 2024 - March 2026

Data Scientist

- Designed and maintained the company chatbot, a multi-agent system with a [Model Context Protocol](#) (MCP) server, improving users' knowledge retrieval and support workflows.
- Contributed to the open-source AI project [mcpadapt](#) on GitHub, strengthening advanced Python patterns and aligning with industry standards.
- Delivered an end-to-end workflow for tabular synthetic data generation using a SOTA tabular foundation model.
- Placed top 5 in the [FDA challenge](#) with a regulatory-compliant RAG model and custom evaluation benchmarks for robustness and traceability.
- Built and deployed reproducible ML pipelines on Azure, supporting research teams with standardized experiment tracking and scalable inference.

Miraitek srl September 2023 - April 2024

Junior Data Scientist

- Built a RAG retrieval agent on AWS that automated document analysis and reduced manual processing time for manufacturing clients.
- Surfaced operational inefficiencies through pattern recognition (7% production-performance improvement) and BI dashboards (Apache Superset + SQL).

SELECTED PROJECTS

[mcpadapt](#) [Open Source](#)

Upstream contributor to an open-source bridge between agent frameworks and Model Context Protocol servers.

[precisionFDA challenge](#)

Top-5 finish with a regulatory-compliant RAG and its own evaluation harness, built for traceability regulators can read.

EDUCATION

MSc in Engineering in Computer Science, [Sapienza University](#) 2023

Minor: Data Science & Machine Learning. Thesis: NLP for Integrity on Social Platforms (with an ex-Meta researcher); +15% accuracy via meta-learning over traditional approaches.

BSc in Software Engineering, [Roma Tre University](#) 2019

Minor: Computer Science.

CERTIFICATIONS

Top 3.6% of HUAWEI Seeds For the Future (2022) - business plan focused on tech innovation, budget and investor engagement.

Additional ML/MLOps/AI certifications on LinkedIn.

SKILLS

AI Agents & LLM Systems: RAG, multi-agent systems, MCP servers, explainable AI outputs. OpenAI, HuggingFace, LangChain, prompt engineering, evals.

ML & Statistical Methods: Supervised/unsupervised learning, tabular synthetic data, statistical analysis components for research workflows. PyTorch, Scikit-Learn, Pandas.

MLOps & Cloud: Reproducible ML pipelines, experiment tracking, scalable inference. Azure ML, AWS, CI/CD, containerization.

Product & Software Engineering: Production Python (FastAPI), React/TypeScript frontend, git, testing, CI quality gates.